Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Ramez Hajj	Email	rhajj@illinois.edu
Institution / Department	Department of Civil & Environmental Engineering, Grainger College of Engineering, University of Illinois Urbana-Champaign			
Address	Newmark Civil Engineering Lab, 205 N Mathews Ave., Urbana, IL 61801			

2. Technology Overview

Title of Proposed Technology	*Binary self-healing retro-reflective pavement markings*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Mechanical - Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Fine chemicals	*Paints/coatings*	*Automobile/Rail Vehicles*	*Other Automobile/Rail Vehicle–related Technologies*
Concept of Proposed Technology	Pavement markings are one of the most critical safety countermeasures in road engineering. The proposed technology involves the replacement of conventional pavement markings, which involve a type of paint with glass beads on top, with binary, self-healing markings which leverage calcium alginate self-healing capsules that our research group has previously developed for asphalt concrete. The novel markings present two major advantages over the current state-of-practice: (1) their binary nature enables better visibility in fog, ice, and other adverse conditions than conventional pavement markings and (2) they are able to repair micro-scale damage in the pavement surface, extending the life of the marking. The proposed technology achieves these goals by emulsifying sodium alginate with calcium chloride and an asphalt healing agent to produce capsules made of calcium alginate which encapsulate the healing agent. To achieve retro-reflectivity, the capsules will now be embedded with micro-prismatic structures, and binary properties are achieved by adding food dye to some capsules to change their color to black, against a typical white or yellow paint color.			

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Rationale for Joint R&D Project	The proposed project requires collaboration with industry to scale the technology and make it suitable for use in roads. Currently, we can produce self-healing capsules at laboratory scale. However, working with industry partners with access to larger-scale production and manufacturing processes will enable our team to advance the technology to the field trial stage by leveraging the connections that industry can provide. Furthermore, industry experts will be necessary to streamline processes, enhance practicality, and refine the technology to optimize chemistry, retro-reflectivity, and other practical considerations. Finally, partnership with partners in South Korea makes sense, as Korea, like the United States, has a network of predominantly asphalt-surfaced roads, with more than 95% of the network being paved and more than 90% of those roads being paved with an asphalt concrete surface. South Korea is also at the forefront of technologies which enable autonomous transportation, which increases the demand and need of the country for novel pavement marking designs.
Keywords	1. safe mobility 2. pavement markings 3. flexible pavements 4. self-healing materials 5. connected and autonomous transportation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	Asphalt materials have been widely used in pavements due to their mechanical properties and high recyclability. One of the challenges of flexible pavements, which are pavements surfaced with asphalt mixtures, is aging because of volatilization, temperature, and ultraviolet light. In recent years, more than 90% of end-of-life asphalt pavements in the U.S. have been recycled (Williams et al., 2020). The growing incorporation of reclaimed asphalt materials increases the concern for fatigue cracking due to their heavily oxidized nature. The intrinsic healing capacity is a function of material properties, aging levels, and temperature. When the healing capacity of aged samples is hindered, extrinsic healing approaches could be considered such as self-healing capsules. Healing capsule incorporation with rejuvenation is an option based on recent studies which observed healing potential improvement for aged asphalt with the addition of rejuvenators (Asadi and Tabatabaee, 2022; Wan et al., 2023). Cracking is a common distress in asphalt pavements, as a combined result of traffic loading, freeze-thaw cycling, temperature change throughout the day, and oxidative aging which occurs through the asphalt concrete layer's life, causing the material to become stiffer and more brittle. Healing is defined as the reversal of the cracking process through the closure of micro-cracks, delaying their formation into larger and more significant cracks which reduce the pavement's lifespan. However, aging caused by oxidation, temperature fluctuations, and solar ultraviolet exposure significantly reduces healing capability (Bhasin et al., 2011). Rejuvenators have been applied and sprayed on pavement surfaces as one method to improve healing capacity of aged asphalt. However, rejuvenators cannot penetrate more than 10-20 mm when sprayed at the surface (Lin et al., 2014). On the other hand, adding rejuvenators at the beginning of the pavement life is a common practice which can improve the entire layer, but acts only once, at the beginning stage and not later in the life as the pavement ages. To alleviate these issues, in recent decades, encapsulated rejuvenation has been proposed as an alternative approach, where rejuvenators are enclosed within soft capsules dispersed within the asphalt matrix (Xu et al., 2018). When cracks develop, these capsules rupture and release the rejuvenator, facilitating localized healing. Tabakovic et al. (2016) proposed encapsulation of			

rejuvenators in calcium alginate fibers which would break and release the rejuvenator as a healing agent as the asphalt matrix cracks. Following a similar formulation, subsequent researchers have designed and optimized alginate-based capsules for various types and scales of asphalt mixtures, including mastic-level mixtures (Al-Mansoori et al., 2019), porous asphalt (Xu et al., 2021), dense-graded mixes (Zhao et al., 2023), and Stone Matrix Asphalt (SMA) (Norambuena-Contreras et al., 2019). However, Li et al. (2021) pointed out that inconsistencies in reported performance improvements, attributing the variability to differences in evaluation criteria, damage and healing conditions, and sample scales.

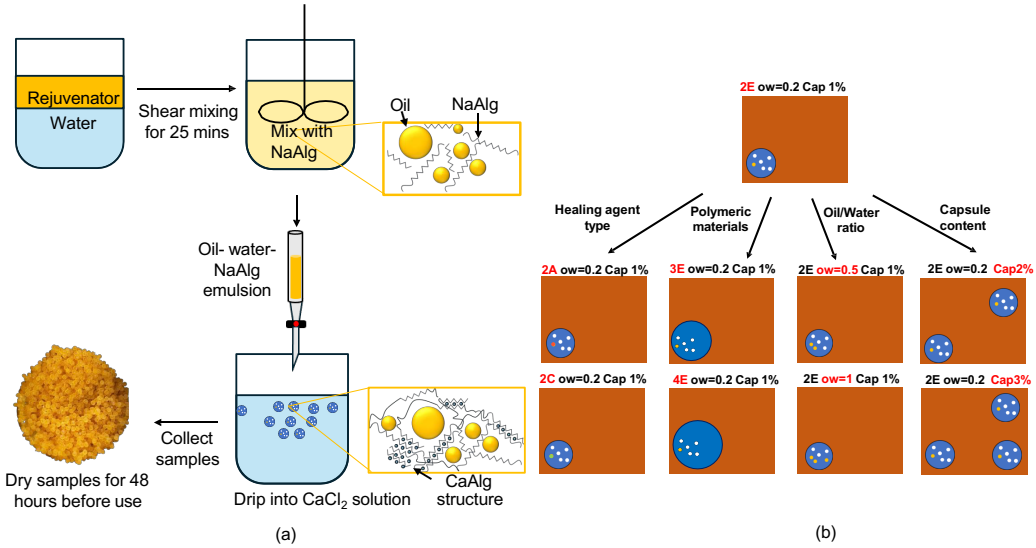

Figure 1 - (a) process for fabricating self-healing capsules and (b) factors evaluated in terms of capsule design in our previous work (Lu et al., 2025).

Therefore, further investigation is needed to optimize capsule design and determine efficiency in pavement by connecting behavior at different length scales. Recently, our team conducted a multi-scale study of capsule design using a partial factorial matrix (Lu et al., 2025) and validated our experimental findings with molecular dynamics simulation (Lu et al., 2024). This study was a first-of-its-kind study which evaluated several different design factors for self-healing alginate capsules, such as the amount of rejuvenator used, the ratio of various components in the formulation, and the amount of capsule used in the mix (Figure 2). The findings indicated that an optimal approach was to utilize various capsules of different sizes, which would break and release the healing agent at different times during the pavement life. The proposed study herein therefore seeks to *implement this type of framework, but now for pavement markings*.

Pavement markings present some similar challenges. They are placed on asphalt concrete, which means that the asphalt concrete underlying them faces similar degradation, especially considering environmental exposure of the markings. Pavement markings are critical traffic control devices which provide continuous, immediate information to road users which informs them of the traffic pattern and prevents crashes. Of critical importance for guiding connected and autonomous vehicles (CAVs) are edge line and lane line markings, which prevent the vehicle from departing the travel lane. Without clearly detectable markings, the chances of run-off-the-road crashes or head-on collisions with opposite direction traffic greatly increase.

Currently, longitudinal pavement markings are largely constructed to meet retro-reflectivity standards using some type of paint (thermoplastic, waterborne epoxy, or others) with glass beads placed on top to reflect light at an appropriate angle to be visible by drivers. Primarily, glass beads in pavement markings work on the mechanism of total reflection of light back towards the light source, which is the measure of the ratio of luminance intensity in candelas over illuminance in lux using a measurement device. This mechanism is different from diffuse reflection, where light scatters in all

directions; specular reflection, where light reflects at the same angle at which it arrives; and mixed reflection, where both diffusive and specular behaviors appear in combination, as shown in Figure 2.

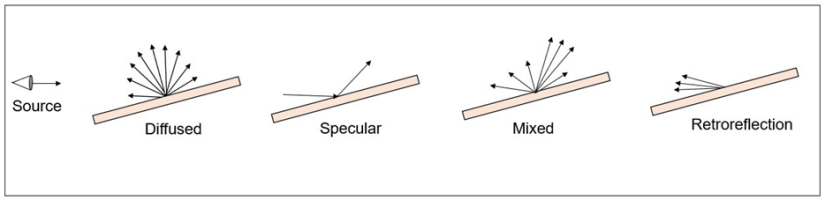

Figure 2 – Reflection types.

Although these marking types are highly effective for human drivers, markings, like the pavement surface, face deterioration over time. Winter maintenance practices are essential during the colder months for ensuring both the operational safety and the longevity of pavements. Such practices include treating the surface with anti-icing and de-icing compounds to prevent and mitigate ice-pack formation. However, these compounds react with glass beads embedded in pavement markings and raise environmental concerns, primarily due to the possible leaching of heavy metals into the surrounding ecosystems (Sandhu 2013). In addition, mechanical processes such as snowplowing can physically damage pavement markings. Mechanical wear accelerates the degradation of retro-reflectivity in pavement markings, which compromises their visibility and durability. Figure 3 shows an example of an edge line marking with substantial degradation including the loss of glass beads on the traffic side and less on the shoulder side. This figure demonstrates how the traffic, and potentially snowplow, loading affects the longevity of the marking by both damage to the painted surface and removal of glass beads.

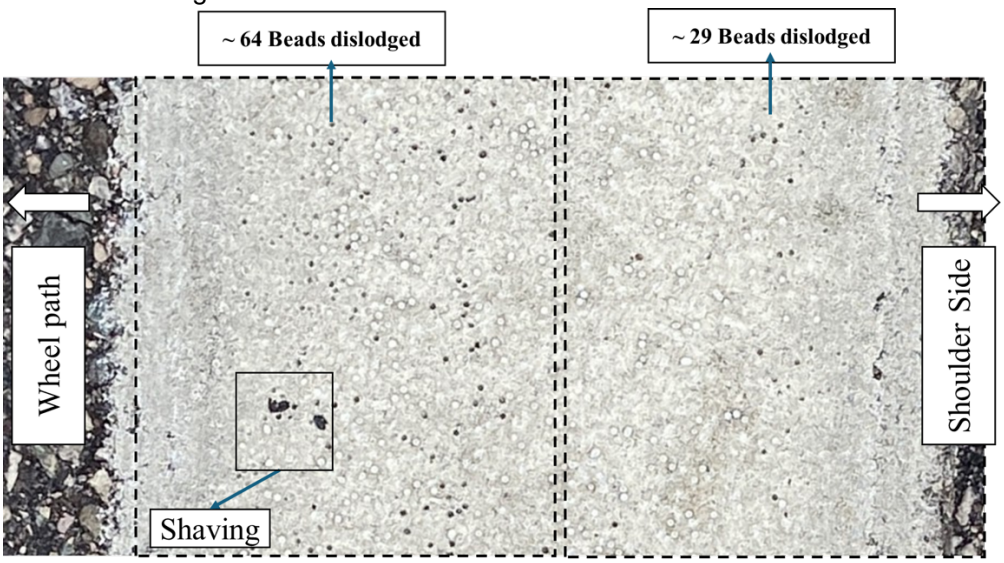

Figure 3 - Marking image taken on MN 95 Woodbury, Minnesota.

Considering these challenges and a general desire to improve the function of pavement markings, several new marking types have been proposed. These marking types include binary pavement markings (Wang et al., 2022), which in the present proposal represent thermally sensitive markings. Based on the above, it is clear that damage to markings is a substantial issue reducing their service life, and that marking visibility is the most critical safety issue, especially in the context of autonomous vehicles, which rely on machine vision or LiDAR to capture where markings are. *The proposed study seeks to therefore utilize binary markings derived from self-healing capsules placed on the marking material, modified with micro-prismatic structures and food dye, to achieve a long-lasting, durable, binary pavement marking.*

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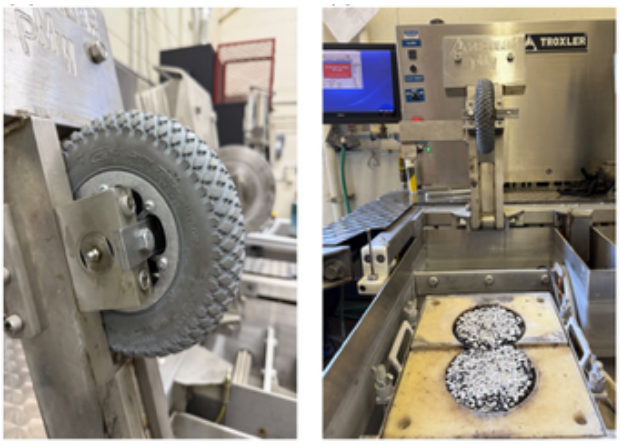
R&D Target Specifications	The research team intends to develop a comprehensive framework for productizing this technology, in conjunction with partners identified through this call. The goal of this project is to produce capsules, at scale, which have both self-healing efficacy and retroreflectivity. The specific metrics will include laboratory evaluation to ensure that the self-healing markings increase overall marking life by at least 20%, ensuring retroreflectivity is comparable to conventional pavement markings, and cost efficiency of the proposed production process, ensuring that the costs are not prohibitive for road agencies.
R&D Plan	**Phase 1 – Seed Phase** The research project will begin with the 6-month seed phase to initiate the collaboration between the research team and the SME. During the seed phase, PI Hajj will visit the SME in-person to develop a strategic plan for the next 2.5 years of the project in collaboration. This will include visits to facilities and meetings with stakeholders and potential partners on the Korean side of the project. Simultaneously, the research team will begin developing a few capsule options in the lab at UIUC, informed by our previous work on self-healing asphalt capsules for asphalt concrete. The research team will conduct preliminary studies using methods detailed below to identify 3-4 candidate capsule types which could be most promising in pavement marking applications. This includes laboratory fabrication of capsules as shown in Figure 1, and testing using an initial test focused on determining what benefits the capsules can provide to a marking placed on an aged pavement surface. Therefore, the research team will artificially age asphalt samples in the oven, then place a pavement marking on top. Micro-prismatic structures of size around 50 microns will be embedded in the capsules at various ratios and using several emulsification methods to ensure that they provide retro-reflectivity and then capsules will be added to the marking mix in place of conventional glass beads. The research team will assess the durability of these capsule-modified markings using a modified Hamburg Wheel Tracking Device instrumented with a pneumatic tire and lighter loading than the conventional test (Figure 4). The device's tire will be locked to simulate braking, and the angle may also be changed to simulate turning. This will allow the researchers to test the degradation of these marking types with no capsules as a control, and with capsules, to identify promising designs for implementation in the Full Funding stage.  Figure 4 - Modified Hamburg Wheel Tracking Device and sample; markings will be applied on top of the sample **Phase 2 – Full Funding** *Task 1 – Performance-based laboratory design of capsules* The research team will design and conduct lab-scale trials for several promising self-healing technologies. First, the research team will develop capsules of several compositions, as performed in our previous works focused on capsule optimization of asphalt concrete (Lu et al., 2024, 2025). This involves emulsifying calcium chloride with sodium alginate and an asphalt recycling agent, which serves as a healing agent herein, to precipitate calcium alginate capsules. The calcium alginate structure is shown in Figure 5.

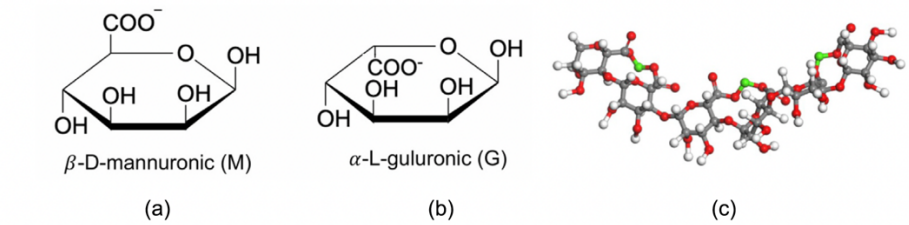


Figure 5 - Calcium alginate structures: (a) 1,4-linked-β-D-mannuroni (M) block, (b) α-L-guluronic (G) block, and (c) structure of calcium alginate in BIOVIA Materials Studio.

As shown in Figure 1, changing various ratios of these materials can be done to tune the overall composition of the capsule. In this project, we will follow our previous experience and use several capsule types in a single marking. In addition, the research team will evaluate several ways to add the retroreflective micro-prismatic structures. These cube-shaped structures utilize three mutually perpendicular surfaces to form the internal corner of a cube, reflecting incoming light back precisely to provide excellent retro-reflectivity. It is believed that these structures are stronger than conventional glass beads, and therefore will improve durability of the marking; however, the research team will also consider use of typical glass beads in the event that this method does not work well. Capsules will also be modified with food dye to make some of them binary, improving their visibility in adverse conditions against a white or yellow pavement marking.

Once the preliminary capsule designs are selected, the research team will begin conducting performance testing on the markings. The performance tests will include several tests which the research team has previously used. The first is the modified Hamburg Wheel Tracking Device test discussed above. This test will involve placing markings and testing in both dry and water-submerged environments. The locked wheel will pass over the marking several times, and the deterioration of the marking will be assessed using a retro-reflectometer and computer vision techniques which our lab is currently developing for assessing marking conditions. The research team will also perform fatigue testing of the specimen using the same device. Although this experiment is not conventionally used for fatigue evaluation, a recent study proposed an effective method for instrumenting the test to produce fatigue cracking under a steel wheel load (Zhang et al., 2021). The research team will apply this method to specimens with the self-healing markings to capture the extension that the capsules provide to the fatigue life of the material underlying the marking.

In addition, the research team will utilize novel cracking methods to assess the impact of capsules in shear. Since shear cracking is the main mechanism expected at the locations where markings are placed, the research team will utilize the ARCAN geometry to assess the pure mode II cracking resistance and mixed-mode cracking in asphalt concrete with self-healing markings. This is critical, as mixed-mode and shear are not typically considered in conventional asphalt cracking tests. Figure 6 shows how we are currently using this test in conjunction with Digital Image Correlation, which will also be critical in the context of this project.

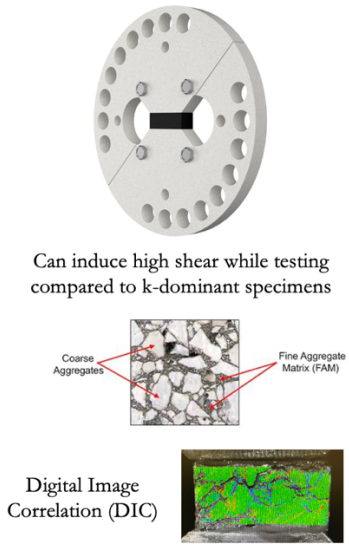


Figure 6 – ARCAN geometry and how we are using it for asphalt concrete.

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	Task 2 – Identification of alternative approaches: induction heating systems To mitigate any risk of the technology not working, the research team will consider alternative approaches to the self-healing via micro-encapsulation of healing agents. The most immediately applicable alternative approach is to use induction-based self-healing. Induction heating can facilitate molecular mobility in asphalt and marking materials, since they have viscoelastic properties, when the materials are embedded with a conductive material. In this case, the self-healing capsule, instead of containing the healing agent, could contain small slag particles, which would then be heated using an inductive system mounted on a vehicle, and could therefore extend the fatigue life of the mix. The research team will fabricate these specimens using 3D printing of the capsules to ensure their durability, and then will perform a similar research matrix to the one proposed in the above Task 1. This will provide a good alternative and multiple feasible options for further implementation. *Task 3 – Scaling and industrialization of healing technology* The major issue which has prevented wide deployment of self-healing asphalt materials is scaling of this technology so that it can be implemented in field settings, which consist of several kilometers of road at a time. Doing this requires large-scale capsule production and ability to blend them with asphalt materials. The latter appears easier in the case of pavement markings, as this can be done directly at a construction site or pre-blended during production of the paint. The UIUC researchers will seek input from the SME on the best procedure by which this can be done. This will require expertise on chemistry of these materials and the production processes. The appropriate partnerships to produce either type of capsule, the calcium alginate healing capsule or the 3D printed inductive capsule, or both, will be formed and production processes will be developed for producing enough capsules to pave 10 km of lane markings in a single day. The research team will also reach out to pavement striping contractors to begin developing a plan and best practices for placing these materials. *Task 4 – Initial Field Trials* The fourth task in this project will focus on conducting preliminary field trials of the self-healing capsule technology. The research team will leverage on-campus test sites at the University of Illinois, which is feasible based on previous approvals from Facilities and Services at UIUC, who have provided support in several proposals for sustainable pavement technology testing on campus as a living lab. Similar studies will be conducted in South Korea, which has an overall similar environment to Urbana-Champaign, but much higher traffic levels in most cities. Therefore, utilization of this technology in several parts of the world will support its translation to practice. The research team will engage road agencies to establish several potential field locations to install the most promising materials from the laboratory study.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The University of Illinois Urbana-Champaign (UIUC) team, led by Prof. Ramez Hajj, will be responsible for conducting laboratory testing during Tasks 1 and 2. Prof. Hajj will visit the SME partner in South Korea at least once per year, and more frequently if needed, to collaborate and integrate the work of both project teams. The UIUC team will also plan and identify potential field sites in Tasks 3 and 4, as well as conduct market research and outreach to contractors and road agencies in the United States.

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Expected R&R of Korean SME Partner	The Korean SME partner will be expected to assist with scaling the technology and developing processes for fabricating self-healing capsules industrially, in collaboration with Prof. Hajj's team. The SME partner will also work with local agencies in South Korea to identify pilot projects and conduct research on the possibilities of introducing the technology there. Finally, other expertise of the SME partner such as on materials chemistry, coatings, additive manufacturing, vehicle autonomy and sensing, or other areas will be greatly appreciated to facilitate the completion of the project.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The outcomes and associated impacts of the project are expected to be as follows: 1. Develop novel self-healing capsules appropriate for use in pavement markings 2. Improve transportation safety by ensuring these capsules result in more durable, reliable pavement markings 3. Improve the life of the underlying pavement by the self-healing mechanism 4. Improve visibility of pavement markings by autonomous vehicles in adverse weather by making them binary 5. Scale the self-healing capsule technology and introduce it to transportation officials and contractors. Completing this project will result in improvements in road safety and better readiness of transportation infrastructure for connected and autonomous vehicles in the United States and South Korea. There is also substantial commercialization potential as, if and when the technology is proven, there is a very substantial market for the product, given pavement markings are ubiquitous, and asphalt surfaces the vast majority of roads in the United States and South Korea.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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